

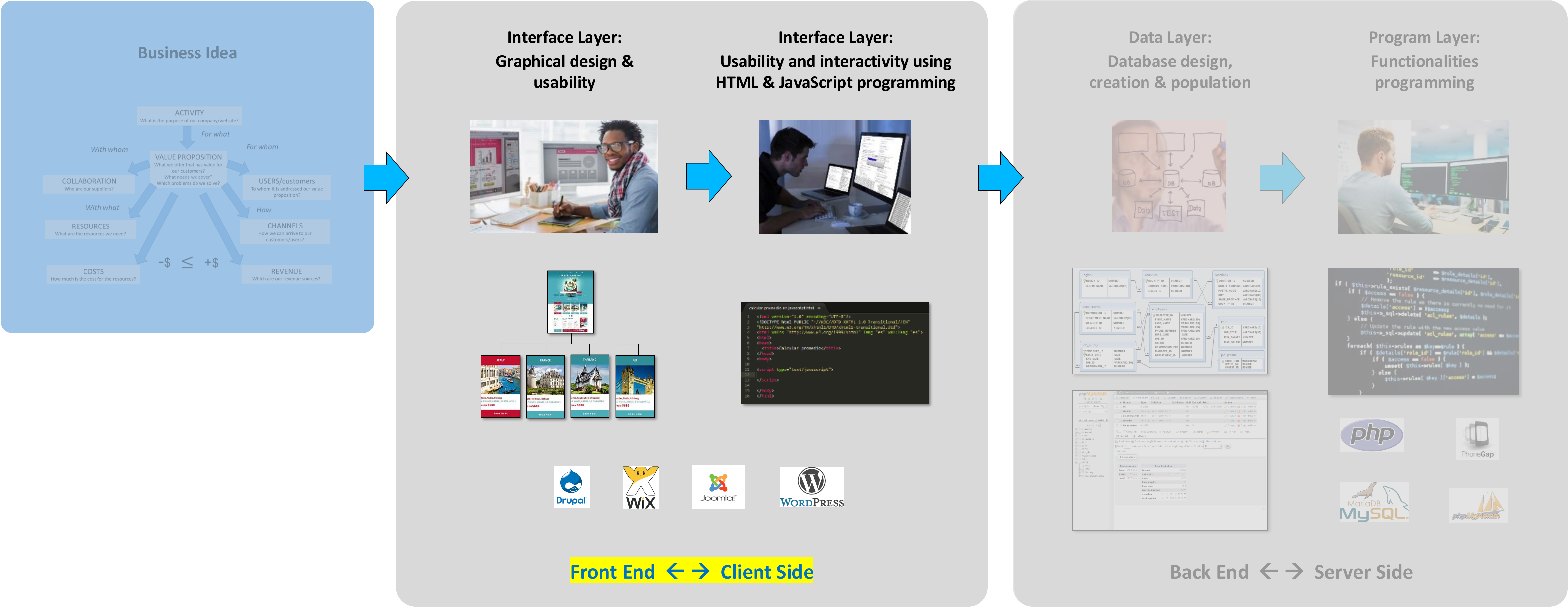
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Development Tools for Web and Mobile Apps

Front End – HTML Forms & JavaScript

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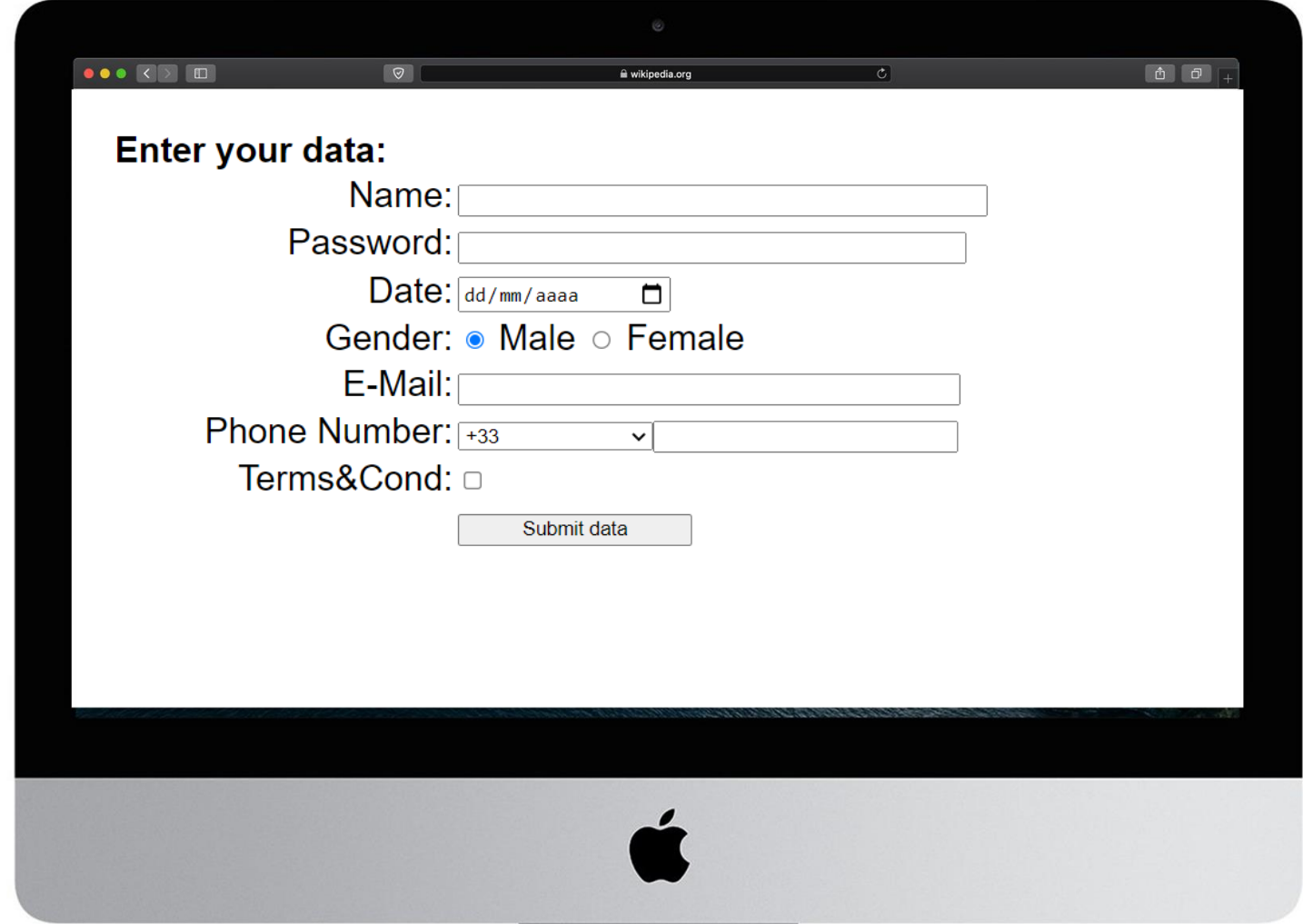
Front End Web development



HTML Forms & JavaScript

Client side

<http://www.mydomain.com/mydata.html>



Enter your data:

Name:

Password:

Date: 

Gender: ☒ Male ☐ Female

E-Mail:

Phone Number:

Terms&Cond: ☐

<http://www.mydomain.com/mydata.html>

Internet

Server side

<http://www.mydomain.com>



welcome.html file

```
<html>
<head>
  <title>My First Web Page</title>
  <script>
    // Javascript code
  </script>
</head>
<body>
  <form name ... >
    ...
  </form>
</body>
</html>
```

JS

JavaScript allows the control of the interaction between the user and the web page elements, in particular with forms controlling data input/output.

It is a programming language, meaning that is a formal language that provides instructions to computers, allowing input/output data control and data manipulation.

JavaScript can be embedded into HTML code and it's interpreted and executed locally in the client side by the browser.

HTML Form

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <meta content="text/html; charset=utf-8">
5      <title>Enter your data</title>
6      <style type="text/css">
7          .style1 {
8              font-size: x-large;
9              font-family: Arial, Helvetica, sans-serif;
10         }
11         .style2 {
12             text-align: right;
13             width: 225px;
14         }
15     </style>
16     <script>
17         // Javascript functions
18     </script>
19 </head>

```

```

20 <body class="style1">
21     <form name="f1">
22         <strong>Enter your data:</strong><br>
23         Name: <input name="name" style="width: 350px;" type="text"><br>
24         Password: <input type="password" id="pass" style="width: 336px;"><br>
25         Date: <input name="bdate" type="date"><br>
26         Gender:
27         <br>
28         <input checked="checked" name="gender" type="radio" value="M"> Male<br>
29         <input name="gender" type="radio" value="F"> Female<br>
30         E-Mail: <input name="email" style="width: 332px;" id="email" type="text"><br>
31         Phone Number:
32         <select name="prefix" style="width: 132px;" id="prefix">
33             <option value="33" selected="selected">+33</option>
34             <option value="34">+34</option>
35             <option value="44">+44</option>
36         </select>
37         <input name="phone" style="width: 198px;" type="text" id="phone"><br>
38         Terms and Cond:<input name="terms" type="checkbox" id="terms" value="1"
39         onclick="check_terms" /><br>
40         <input name="Submit1" style="width: 158px;" type="button" value="Submit data"
41         onclick="check_email">
42     </form>
43 </body>
44 </html>

```

HTML Form

HTML FORMS - Elements Summary

Type	Attributes (example)	Generic Description
<form>	name="f1"	Container that groups input elements and allows data submission.
<input type="text">	name="name"	Single-line text field for user input.
<input type="password">	name="password"	Text field that hides the characters entered (for passwords).
<input type="date">	name="bdate"	Input field with a calendar UI for selecting dates.
<input type="radio">	name="gender"	Radio button used to select one option from a group.
<input type="checkbox">	name="terms"	Checkbox used to select or confirm an option (on/off).
<select>	name="prefix"	Dropdown menu that lets users choose one option from a list.
<option>	value="33"	Represents one item inside a <select> dropdown.
<input type="button">	value="Submit data"	Button that performs an action when clicked (does not submit form by default).
id	id="xdate"	Unique identifier for an HTML element (used in JavaScript or CSS).
name	name="email"	Identifies the form data key when submitted; also used in JavaScript or backend.
required	(no value needed)	Specifies that the input must be filled out before submitting the form.
onchange	onchange="..."	JavaScript event that runs when the value of an element changes.
onclick	onclick="..."	JavaScript event that runs when an element (e.g., button) is clicked.

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Event management for HTML Forms input/output control

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4  ...
5  <script>
6  function check_email() {
7      var email = document.getElementById("email").value; // Gets the email from the email field
8      if (email.indexOf("@") == -1) { // check if there is an @ in the mail. If not, the function returns -1.
9          //If there is an @ the function returns the position of the @ in the mail. The first cha
10         alert("The email must include an @");
11     }
12 }
13 </script>
14 </head>
15 <body>
16 <form name="f1" class="style1">
17     <strong>Enter your data:<br /> </strong>
18     Name: <input name="name" style="width: 350px;" type="text" />
19     Password: <input type="password" id="pass" style="width: 336px;" required onchange="pass_long()"/>
20     Date: <input name="bdate" type="date"/>
21     Gender:
22     <input checked="checked" name="gender" type="radio" value="M" /> Male
23     <input name="gender" type="radio" value="F" /> Female
24     E-Mail: <input name="email" style="width: 332px;" id="email" type="text" required onchange="check_email()"/>
25     Phone Number:
26     <select name="prefix" style="width: 132px;" id="prefix">
27         <option value="33" selected="selected">+33</option>
28         <option value="34">+34</option>
29         <option value="44">+44</option>
30     </select>
31     <input name="phone" style="width: 198px;" type="text" id="phone" required onchange="check_phone()" />
32     Terms&Cond: <input name="terms" type="checkbox" id="terms" value="1" />
33     <input name="Submit1" style="width: 158px;" type="button" value="Submit data" onclick="check_terms()" />
34 </form>
35 </body>
36 </html>
```

Example:

Onchange effect on email field

When the content of the field with **name email** is changed by the user, when leaving the field (with tab or clicking with the mouse in any other different element) **onchange event** will be activated, with the automatic execution of the **JavaScript check_email() function** associated.

Typically the purpose of the JavaScript function is to perform some checks to the data introduced in the associated field.

Some events associated with HTML Forms

```
<input type="text" id="myFieldId" onblur = "myJavascriptFunction()">
```

It is recommended that all HTML elements have an *id* defined. An *id* is a name that uniquely identifies this element throughout the HTML page. The *id* will be used in the Javascript function to access the element, request its state (values) and apply the necessary changes (change values).

Onblur is one possible event associated with field formation, but there are others. The most commonly used are:

onblur	The user is leaving the current form field. The user pressed tab to go to the next form field or clicked the mouse on a different form field.
onchange	When the value of the field changes
onclick	The user clicks on an HTML element
onmouseover	The user moves the mouse over an HTML element
onmouseout	The user moves the mouse away from an HTML element.
onkeydown	User presses a key on the keyboard
onload	The browser has just loaded the HTML page

JavaScript conditions syntax

if (a == b)	condition is true if a and b have same values
if (a != b)	condition is true if a and b have different values
if (a < b)	condition is true if value of a is less than the value of b
if (a > b)	condition is true if value of a is greater than the value of b
if (a <= b)	condition is true if value of a is less or equal than the value of b
if (a >= b)	condition is true if value of a is greater or equal than the value of b
if ((a>0) (b>0))	OR connection between two conditions Final condition is true if any f the partial conditions is true
if ((a>0) && (b>0))	AND connection between two conditions Final condition is true only if both partial conditions are true

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Form Example: Functions pass_long() and check_terms()

```
17 function pass_long() {
18     var passw = document.getElementById("pass").value; //Gets de password from the password field
19     if (passw.length < 8) { // checks if the password is less than 8 characters long
20         alert("The password must have a minimum of 8 characters.");
21         document.getElementById("pass").value = ""; //clears the password text field
22         document.getElementById("pass").focus(); // place the cursor in the password field
23     }
24 }
25
26 function check_terms() {
27     if (document.getElementById("terms").checked == false) { //check whether the checkbox is checked
28         alert("You must accept the terms");
29     }
30     else {
31         alert("Form sent");
32     }
33 }
```

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Form Example: Functions check_email() and check_phone()

```
35 function check_email() {
36     var email = document.getElementById("email").value; // Gets the email from the email field
37     if (email.indexOf("@") == -1) { // check if there is an @ in the mail. If not, the function returns -1.
38         //If there is an @ the function returns the position of the @ in the mail. The first cha
39         alert("The email must include an @");
40     }
41 }
42
43 function check_phone() {
44     var xprefix = document.getElementById("prefix").value; // Gets the prefix & phone number
45     var xphone = document.getElementById("phone").value;
46
47     if (xprefix == 34) // check if it is a Spanish mobile phone
48     {
49         if (xphone.indexOf("6") != 0 ) // In case it is a Spanish mobile phone, check if the first number (position 0) is not
50         {
51             alert ("mobile phones in Spain must start with a 6"); // Display the error message
52         }
53     }
54 }
```

A dark blue diagonal banner runs across the top of the image, sloping downwards from left to right.

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